



Research Plan on the Sabbatical

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Background:

Open 3D-SDI and Sustainability

Global Expansion and Applications

- Open 3D geospatial data supports smart cities, urban planning, disaster management, and civic engagement worldwide.

National Initiatives and Challenges: for decentralization SDI

- Japan's PLATEAU and the Netherlands' 3DBAG show government investment in open 3D data with sustainability challenges.

Socio-Technical System

- Open 3D data functions as a socio-technical system involving governments, academia, private sectors, and citizens.

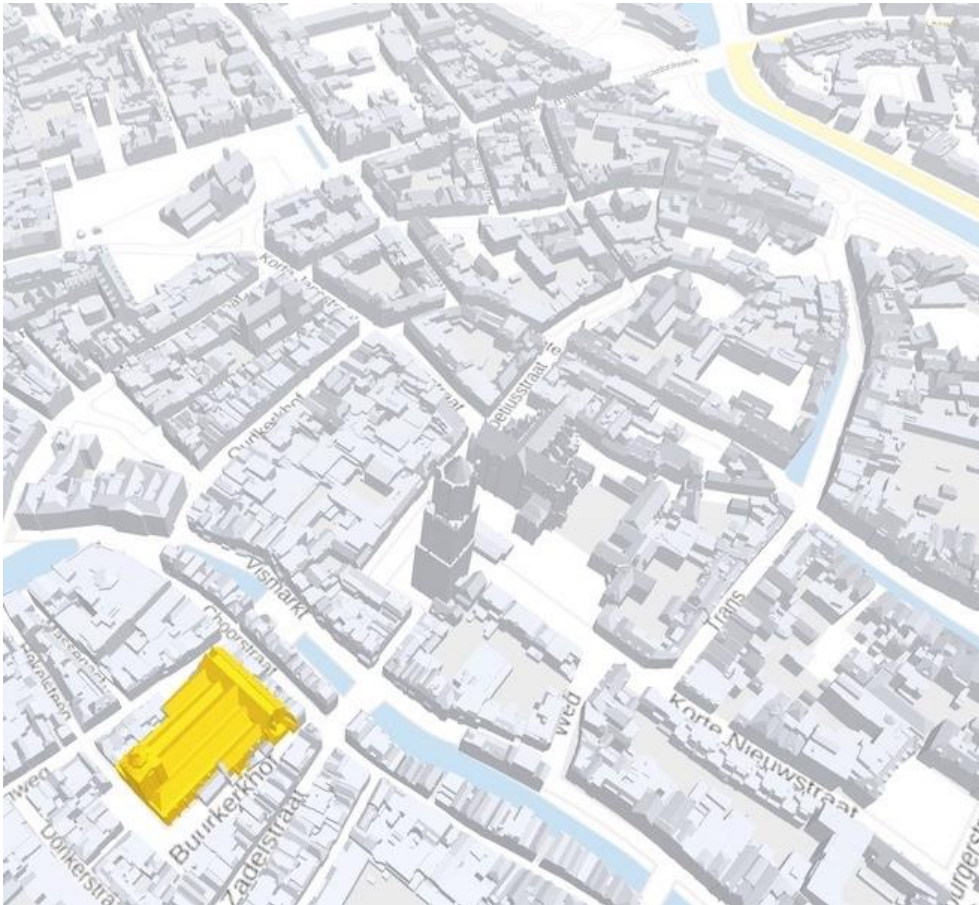
Context-Sensitive Sustainability

- Sustainability requires understanding how data infrastructures are maintained, taught, and used beyond technical aspects.

Research Question

- **“How can open 3D geospatial data ecosystems achieve long-term sustainability SDI?”**
 - ✓ RQ1: What is the requirement for "3DGIS Data Commons (infrastructure)" ?
 - ✓ RQ2: Citizen Engagements for 3D data Maintenance and Usage
 - ✓ RQ3: GIS education and knowledge transfer for 3DGIS

RQ1: What is the requirement for "3D GIS Data Commons"



Development and Institution

3DBAG has been developed and institutionalized over time involving multiple actors ensuring sustainability. How will the budget and organizational structure be maintained?

Roles of Stakeholders: de-centralization

Government agencies, academic institutions, and other stakeholders share responsibilities for maintaining 3DBAG.

Comparative Governance Models

3DBAG's governance contrasts with Japan's PLATEAU project in terms of scaling and institutional approaches.

Research and Knowledge Access

TU Delft researchers engage in discussions to reveal structural factors for infrastructure's long-term viability.

RQ2: Citizen Engagements for 3D data Maintenance and Usage



3D GIS in Citizen Engagements

3D GIS datasets enable public participation and decision-making through immersive 3D visualizations. Furthermore, integration with human-centered VGI data is also expected.

Local Case Studies in Netherlands and Japan

Dutch cases show practical use of 3D GIS (Amsterdam?) in urban planning, environment, and civic tech initiatives.

Sustainability and Societal Relevance

Spatial Data infrastructures must support meaningful civic use to avoid underutilization despite technical quality.

RQ3: Education and Knowledge Transfer



3D geoinformation

Education

MSc Geomatics courses we are teaching

- GEO1002: Geographical information systems (GIS) and cartography
- GEO1004: 3D modelling of the built environment [🏠 \(open course\)](#)
- GEO1015: Digital terrain modelling [🏠 \(open course\)](#)
- GEO1016: Photogrammetry and 3D computer vision [🏠 \(open course\)](#)
- GEOIT1501: Synthesis project [🏠](#)
- GEO2011: Thesis preparation
- GEO2021: Thesis [🏠](#)
- GEO5010: Research assignment [🏠](#)
- GEO5014: Geomatics as support for energy applications
- GEO5015: Modelling wind and dispersion in urban environments
- GEO5016: Geomatics in Practice (Internship) [🏠 \(open course\)](#)
- GEO5017: Machine Learning for the Built Environment [🏠 \(open course\)](#)
- GEO5019: Geomatics Studio (special topic) [🏠 \(open course\)](#)

Role of Education in 3D GIS (and Geo-AI) ?

Education shapes future professionals' skills and values in sustaining open 3D geospatial data ecosystems.

TU Delft's Educational Model

TU Delft integrates open data, open-source tools, and real-world datasets into GIS curricula effectively. What about studying Geo-AI?

Localization for Japan

Adapting educational materials for Japanese universities considers **undergraduate student** differences.

We are discussing the **future vision** for Project PLATEAU next 5 years.

今後取り組むべき重要項目

- PLATEAUビジョン2026では、①データ整備・更新の効率化、②サービス実装、③国外への都市デジタルツイン展開を前述の主な課題の解決をはじめ、Project PLATEAUの目指す姿の実現に向けた**重要項目として位置づけ**

今後取り組むべき重要項目

1

データ整備・更新 の効率化

Data maintenance & update

- モデルデータの整備・更新コストの低減に向けた技術開発と仕様の見直しにより、ニーズに即したモデルの新規整備都市拡大と、既存整備都市のモデル更新サイクルの確立を図る
- ニーズに即したデータ整備の促進
 - 整備・更新の効率化のための先進技術の開発・導入
 - 都市デジタルツイン※3の整備拡充に向けた都市に関連する多様なデータ群の活用

2

サービス実装

Implementation as a service

- 3D都市モデルをまちづくり分野において定常的に活用するとともに、周辺領域においても課題解決のツールとして活用し、さらに、まちづくりを超えた多様な分野における利活用の拡大を図る
- 都市計画・まちづくり実務や、手続き・管理などの行政プロセスの高度化・効率化
 - 社会課題解決に資するビジネス創発
 - 広範な利用者層に向けたさらなる価値創出

3

国外への都市 デジタルツイン展開

Expanding Internationalization

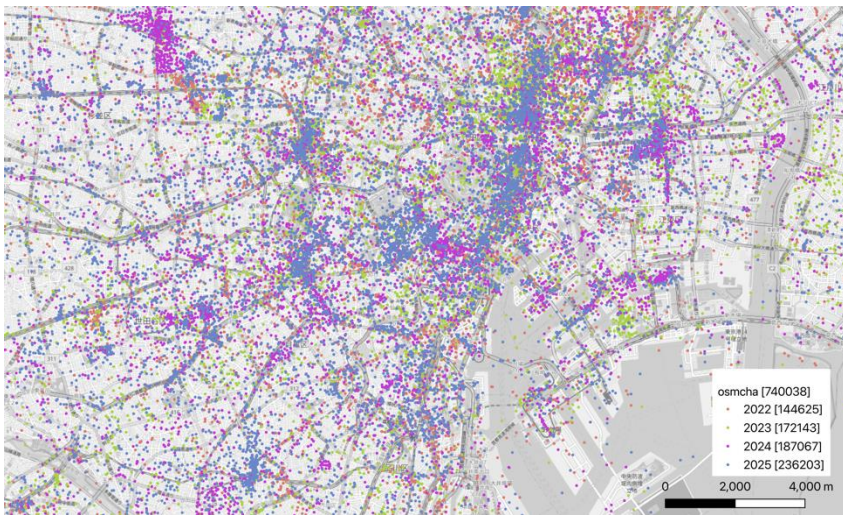
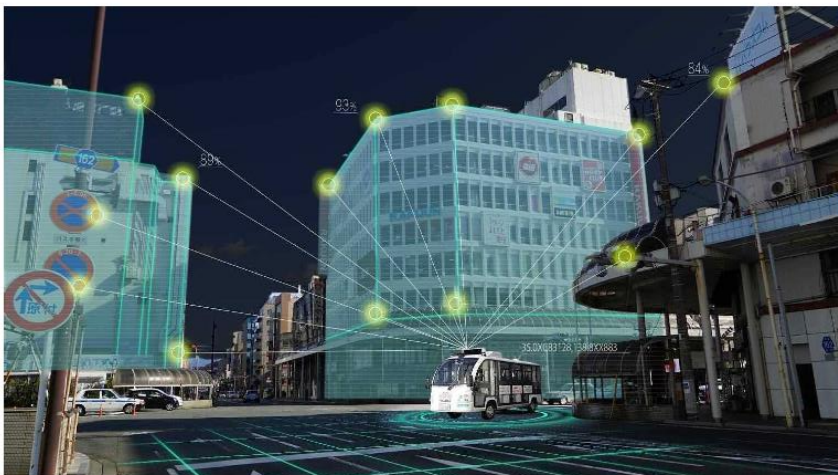
- 国外における都市デジタルツインを活用した課題解決を促進するとともに、国内の取組への海外市場における成果の還元を図る
- 国際社会における我が国取組の認知度向上・主流化
 - 海外の都市開発施策との連動

Other Questions on the Utilization of 3D City Models (in the Netherland and other Europe countries)

Scope and Diversity of Use Cases	What types of application domains currently make effective use of 3D city models? Which sectors (e.g., cultural heritage, education, social sciences, mobility studies) remain underrepresented or insufficiently engaged?
High-LOD Modeling for Cultural and Historical Assets	To what extent are detailed 3D models (LoD2 or higher) developed for World Heritage sites and historically significant structures (e.g., windmills in Kinderdijk)? Are such models actively used for landscape simulation, heritage conservation, or public communication?
Collaboration with Cultural Institutions (Museums)	Given the existence of over 500 museums in the Netherlands, are there systematic initiatives linking 3D city models with museums, archives, or other cultural institutions? How do these collaborations differ from practices observed in other countries?
Integration with Mobility and Spatiotemporal Data	Are there empirical studies that integrate 3D city models with human mobility data (e.g., pedestrians, cyclists, public transport), enabling 4D GIS or time-geographic analyses? What methodological and institutional challenges limit such integration?
Impact of AI and Automation	How is the role of AI perceived and implemented in the 3D GIS ecosystem, particularly with respect to automated model generation, simulation, and scenario analysis? Does AI enhance sustainability or introduce new dependencies and risks?
Implications for GIS Education	What new educational impacts does 3D GIS bring to GIS education, especially for non-native English-speaking students? What challenges and pedagogical innovations (e.g., communities such as geogeeks) have emerged to support inclusive learning?

Research Agenda and On-going Works

3D都市モデルを活用した自動運転車両の自己位置推定
技術検証レポート



Spatiotemporal Analysis of OSM Editing Activity in Japan (FOSS4G 2026 academic track)

Long-term analysis of **VGI activity** provides OSM data for international comparison and FOSS4G2026 conference presentations.

Use-Case Evaluation for PLATEAU (on-going)

Publication in journal following the development of PLATEAU use-case documents through **text mining** and **data coverage** analysis , and its evaluation of sustainability governance.

3D Data Usage Evaluation International Comparative Studies (with Baba-san?)

A comparison of 3D data coverage, types (building, road, trees etc...), use cases, governance between **Europe (e.g., 3D BAG)** and **Japan**.

(Side-works)

1. Operation of FOSS4G Hiroshima 2026 Academic
2. Preparing for publication of a Japanese textbook on GIS and civic tech for undergraduate students.

Task and Schedule (tentative; flexible)

- **April – May 2026:**
 - Comprehensive literature review on 3D city model and Open Data initiatives, with the Netherlands and European countries
 - Understanding Fundamental knowledge on 3DBAG data structure, standards, and update workflows
 - A review of Bastiaan et al. papers are also needed.
 - Networking for Dutch and Europe case studies (researchers, practitioners, public agencies)
 - If you know of any relevant case studies or people, please let me know.
 - **Preparation of guest lecture of PLATEAU (in May?)**
- **June – July 2026:** (RSA@Sweden, ISPRS@Tronto)
 - Spatiotemporal analysis of OpenStreetMap editing activity in Japan/Netherland (for ISPRS Archives)
 - Continued networking and identification of concrete Dutch case
 - Survey of 3D city model usage/application examples (basically desktop survey: Web and GitHub...)
- **August – September 2026:** (Hiroshima and Tokyo)
 - Participation in and operation-related contributions to **FOSS4G Hiroshima 2026**
 - Fieldwork and interviews in Japan (**23 Aug – 14 Sep**), complementing European cases
 - Preparation of a proposal “KAKENHI (JSPS) Japanese grant” focused on 3DGIS and VGI

Task and Schedule (tentative; flexible)

- **October – November 2026:**

- Evaluation of PLATEAU use cases and sustainability through text mining and quantitative analysis
- Journal manuscript preparation focusing on PLATEAU sustainability
- In-depth case studies in the Netherlands and surrounding regions (citizen engagement, governance, education)

- **December 2026 – January 2027:**

- Integrated comparative analysis across datasets, case studies, and educational practices
- Consolidation of findings into coherent analytical frameworks

- **February – March 2027:**

- Final writing and submission of journal articles
- Documentation of outcomes and future collaboration roadmap with TU Delft and partners